1. Core Embedded Components (For the Bat Prototype)

These are the electronic parts that will form the "brain" of your smart bat. It's wise to buy at least two of each key component in case one gets damaged during testing.

- ☑ **Microcontroller (MCU):**
  - **Item:** ESP32-WROOM-32 Development Board
  - **Quantity:** 1
  - **Why:** This is the main processor. The development board version includes a USB port, which makes programming and testing extremely easy.
- ☑ **Motion Sensor (IMU):**
  - **Item:** BNO055 9-Axis Absolute Orientation Sensor Module
  - **Quantity:** 1
  - **Why:** This is the best choice for accurately tracking the 3D swing path and speed. It has an internal processor that handles complex sensor fusion, saving you a lot of coding effort.
- If not available go for MPU-9250 or ICM-20948
- ☑ **Impact Sensors:**
  - **Item:** Piezoelectric Disc Sensors (27mm diameter is a common size)
  - **Quantity:** Pack of 10
  - **Why:** You'll need 3-4 per bat to detect the ball's impact location. They are inexpensive, so buying a pack is economical and provides spares.
- ☑ **Power System:**
  - **Item 1:** Lithium-Polymer (LiPo) Battery
    - **Specs:** 3.7V, 400mAh to 500mAh capacity, with a 2-pin JST-PH connector attached.
    - **Quantity:** 2
  - **Item 2:** TP4056 Li-Ion Charger Module
    - **Specs:** Make sure to get the version with **USB-C port** and **battery protection circuit**.
    - **Quantity:** 1

2. Prototyping & Development Tools

These are the essential tools and accessories for your workbench to connect, program, and debug the electronics.

- ☑ **Wires & Connectors:**
  - **Item 1:** Jumper Wires (get a pack with 1) Male-Male, 2)Male-Female, and 3) Female-Female types).
  - **Item 2:** Thin single-core hook-up wire (e.g., 22 AWG) for making more permanent connections later. - 2 meter

☑ **Essential Tools:**

- **Item 1:** Electrical Tape and double sided tape